

Mathematical Derivation of SER–SNR Relationships for Rydberg Atomic Receivers

Conventional, LO-Free and LO-Dressed Receivers

Based on the paper
“Harnessing Rydberg Atomic Receivers: From Quantum Physics to
Wireless Communications”
and Proakis, *Digital Communications*, 4th Edition

M.Tech Thesis Simulation Reference

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1 Purpose and Overall Mathematical Structure

The objective is to establish the mathematical framework required to reproduce the Symbol Error Rate (SER) versus Signal-to-Noise Ratio (SNR) curves associated with Fig. 9 of the reference paper.

The paper considers three receiver configurations:

1. Conventional RF receiver,
2. LO-free Rydberg atomic receiver,
3. LO-dressed Rydberg atomic receiver.

The paper states that the SER of PAM is calculated according to Proakis, 4th Edition, p. 266. The equivalent Proakis expression used here corresponds to Eq. (5-2-45) (with $\gamma = E_s/N_0$); the paper's citation is conventionally written as Eq. (5.246). The QAM SER is evaluated according to Proakis, p. 278, Eq. (5.279).

The fundamental communication-theoretic structure is

$$\boxed{\text{Receiver model} \longrightarrow \text{effective SNR} \longrightarrow \text{Proakis SER.}} \quad (1)$$

For the Rydberg receivers, an additional receiver-response layer is required:

$$\boxed{\text{RF signal} \longrightarrow E_{\text{RF}} \longrightarrow \Omega_{\text{RF}} \longrightarrow \text{Rydberg response} \longrightarrow \text{effective SNR} \longrightarrow \text{SER.}} \quad (2)$$

Thus,

Conventional: $SNR \rightarrow M\text{-QAM} \rightarrow SER$ LO-free: $SNR \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{AT condition} \rightarrow M\text{-PAM} \rightarrow SER$ LO-dressed: $SNR \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{linear/nonlinear response} \rightarrow M\text{-QAM} \rightarrow SER$
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The classical SER equations are supplied by Proakis. The receiver-specific Rydberg operating and distortion behavior is supplied by the atomic receiver model and, where necessary, by the QuTiP quantum-system simulation.

2 Common Wireless Model

The received RF signal at the vapor-cell location is modeled as

$$\boxed{y = \sqrt{P_{\text{Rx}}} hx + n_{\text{BGN}}.} \quad (3)$$

Here x is the transmitted complex symbol, h is the complex wireless channel coefficient, P_{Rx} is the received RF power, and n_{BGN} is the background noise.

The paper defines

$$\boxed{P_{\text{Rx}} = |E_{\text{RF}}|^2.} \quad (4)$$

Therefore, in terms of field magnitude,

$$\boxed{|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}.} \quad (5)$$

The free-space received-power expression used in the paper is

$$P_{\text{Rx}} = \frac{P_{\text{Tx}} G_{\text{Tx}}}{4\pi d_{\text{Tx-Rx}}^2} \eta_0. \quad (6)$$

For the numerical reconstruction, all quantities must be expressed in consistent SI units. In particular, antenna gains quoted in dBi must be converted to linear gain before numerical substitution:

$$G_{\text{linear}} = 10^{G_{\text{dBi}}/10}. \quad (7)$$

The communication chain is therefore

$$P_{\text{Tx}} \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow \Omega_{\text{RF}}. \quad (8)$$

3 Classical SER Equations from Proakis

3.1 M-PAM SER

For M -ary PAM, the SER expression used for the Fig. 9 reconstruction is

$$SER_{\text{PAM}} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6}{M^2-1} \frac{E_s}{N_0}}\right). \quad (9)$$

Define

$$\gamma = \frac{E_s}{N_0}. \quad (10)$$

Then

$$SER_{\text{PAM}}(\gamma, M) = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\gamma}{M^2-1}}\right). \quad (11)$$

The Gaussian Q -function is

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right). \quad (12)$$

3.2 Square M-QAM SER

For square M -QAM,

$$SER_{\text{QAM}} = 1 - \left[1 - 2\left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3E_s}{(M-1)N_0}}\right)\right]^2. \quad (13)$$

Using $\gamma = E_s/N_0$,

$$SER_{\text{QAM}}(\gamma, M) = 1 - \left[1 - 2\left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3\gamma}{M-1}}\right)\right]^2. \quad (14)$$

For square QAM, the I and Q components have the same probability of correct detection. If $P_{c,I} = P_{c,Q}$, then

$$P_c = P_{c,I}P_{c,Q} = P_{c,I}^2$$

and

$$SER = 1 - P_c,$$

which gives Eq. (12).

For a plotted SNR axis expressed in dB,

$$\boxed{\gamma = 10^{SNR_{dB}/10}}. \quad (15)$$

This mapping is the standard conversion used when the plotted SNR is identified with the E_s/N_0 -type argument of the Proakis expression.

4 PART A – Conventional RF Receiver

4.1 Conventional Received Signal

The conventional receiver is modeled as

$$\boxed{y = \sqrt{P_c}hx + n_{\text{Conv}}}. \quad (16)$$

The aggregate receiver noise is modeled as circularly symmetric complex Gaussian noise:

$$\boxed{n_{\text{Conv}} \sim \mathcal{CN}(0, \sigma_{\text{Conv}}^2)}. \quad (17)$$

Therefore,

$$\boxed{\mathbb{E}[n_{\text{Conv}}] = 0, \quad \mathbb{E}[|n_{\text{Conv}}|^2] = \sigma_{\text{Conv}}^2}. \quad (18)$$

For

$$n_{\text{Conv}} = n_I + jn_Q,$$

the real and imaginary components satisfy

$$n_I, n_Q \sim \mathcal{N}\left(0, \frac{\sigma_{\text{Conv}}^2}{2}\right).$$

4.2 Conventional Noise Model

The conventional receiver noise variance is modeled using the literal noise expression of the reference model:

$$\boxed{\sigma_{\text{Conv}}^2 = G_{\text{Rx}}G_{\text{LNA}}\sigma_{\text{BGN}}^2 + 4k_BTB}. \quad (19)$$

Here G_{Rx} is the receiver antenna gain, G_{LNA} is the LNA gain, σ_{BGN}^2 is the background-noise power, k_B is Boltzmann's constant, T is temperature, and B is bandwidth.

For the conventional receiver, the received signal power is written using the effective receive-aperture expression

$$\boxed{P_c = \left[\frac{P_{\text{Tx}}G_{\text{Tx}}}{4\pi d_{\text{Tx-Rx}}^2} \right] \left[\frac{\lambda_{\text{RF}}^2}{4\pi} \right]}. \quad (20)$$

The first factor is the incident RF power flux density at the receiver location, while the second factor is the effective aperture associated with the receive antenna. The conventional SNR is therefore

$$\gamma_{\text{Conv}} = \frac{P_c |h|^2}{G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B}. \quad (21)$$

This is the literal gain/noise form used for the conventional receiver. No input-referred rewrite is introduced here, because the signal and noise expressions are retained in the same receiver-output convention.

4.3 Conventional Received Signal Power

The conventional receiver uses the incident power flux density multiplied by the receive antenna effective aperture:

$$P_c = \left[\frac{P_{\text{Tx}} G_{\text{Tx}}}{4\pi d_{\text{Tx-Rx}}^2} \right] \left[\frac{\lambda_{\text{RF}}^2}{4\pi} \right]. \quad (22)$$

The RF wavelength used in the effective-aperture term is

$$\lambda_{\text{RF}} = \frac{c}{f_{\text{RF}}}. \quad (23)$$

Thus P_c is the received conventional RF signal power used in the conventional SNR calculation.

4.4 Conventional Signal Power

The conventional received signal is modeled as

$$y_s = \sqrt{P_c} h x. \quad (24)$$

For

$$\mathbb{E}[|x|^2] = 1, \quad \mathbb{E}[|h|^2] = 1,$$

the average received signal power is

$$P_{\text{signal}} = P_c. \quad (25)$$

The channel realization $|h|^2$ is retained explicitly in the instantaneous SNR expression.

4.5 Conventional SNR

Using the conventional received power P_c and the literal receiver noise variance,

$$\gamma_{\text{Conv}} = \frac{P_c |h|^2}{G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B}. \quad (26)$$

For the unit-average channel convention,

$$\mathbb{E}[|h|^2] = 1,$$

the average SNR is obtained by setting $|h|^2$ to its mean value.

4.6 Conventional Modulation

For Fig. 9(a), the conventional receiver uses QAM:

$$\boxed{\text{Conventional} \longrightarrow M\text{-QAM.}} \quad (27)$$

Hence

$$\boxed{SER_{\text{Conv}} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3\gamma_{\text{Conv}}}{M-1}} \right) \right]^2.} \quad (28)$$

4.7 Direct SER–SNR Relationship

If the horizontal axis is SNR_{dB} ,

$$\gamma_{\text{Conv}} = 10^{SNR_{\text{dB}}/10},$$

when the plotted SNR is treated directly as the Proakis SNR argument.

Therefore,

$$\boxed{SER_{\text{Conv}}(SNR_{\text{dB}}, M) = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3 \cdot 10^{SNR_{\text{dB}}/10}}{M-1}} \right) \right]^2.} \quad (29)$$

4.8 Behavior of the Conventional Curve

At fixed M ,

$$SNR \uparrow \implies \gamma \uparrow \implies Q(\cdot) \downarrow \implies SER \downarrow.$$

At fixed SNR,

$$M \uparrow \implies \text{constellation spacing} \downarrow \implies SER \uparrow.$$

Thus,

$$\boxed{SNR \uparrow \Rightarrow SER \downarrow, \quad M \uparrow \Rightarrow SER \uparrow.} \quad (30)$$

5 PART B – LO-Free Rydberg Atomic Receiver

5.1 LO-Free Receiver Signal Model

The LO-free receiver is modeled as

$$\boxed{z = \frac{\wp_{\text{RF}}}{\hbar} \left| \sqrt{P_{\text{Rx}}} hx + n_{\text{Ry}} \right|.} \quad (31)$$

The RF transition dipole moment is retained as $\wp_{\text{RF}}$, without an absolute-value operator, following the notation of the reference paper.

The magnitude operation is fundamental because the LO-free architecture reads the amplitude associated with the AT splitting rather than directly recovering the RF phase. Consequently,

$$\boxed{\text{LO-free} \longrightarrow M\text{-PAM.}} \quad (32)$$

5.2 RF Electric Field

The received RF power is

$$P_{\text{Rx}} = |E_{\text{RF}}|^2. \quad (33)$$

Hence the RF-field magnitude is

$$|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}. \quad (34)$$

5.3 RF Rabi Frequency

The RF field couples the Rydberg states through the RF transition dipole moment:

$$\Omega_{\text{RF}} = \frac{\wp_{\text{RF}} E_{\text{RF}}}{\hbar}. \quad (35)$$

In magnitude,

$$|\Omega_{\text{RF}}| = \frac{|\wp_{\text{RF}}| |E_{\text{RF}}|}{\hbar}.$$

Using the received-power relation,

$$\Omega_{\text{RF}} = \frac{\wp_{\text{RF}}}{\hbar} \sqrt{P_{\text{Rx}}} \quad (36)$$

up to the field-sign convention.

Thus,

$$P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}|.$$

The quantity Ω_{RF} is an angular frequency. The frequency plotted in MHz is $\Omega_{\text{RF}}/(2\pi)$.

5.4 Autler–Townes Splitting

When the coupling laser is scanned, the paper gives the AT splitting as

$$\Delta\nu_{\text{AT}} = \frac{\Omega_{\text{RF}}}{2\pi}. \quad (37)$$

The supplementary derivation gives the extrema at

$$\Delta_{c,1} = 0, \quad \Delta_{c,2} = \frac{\Omega_{\text{RF}}}{2}, \quad \Delta_{c,3} = -\frac{\Omega_{\text{RF}}}{2}. \quad (38)$$

Therefore the observable splitting interval is

$$|\Delta_{c,2} - \Delta_{c,3}| = \Omega_{\text{RF}}. \quad (39)$$

This establishes the physical basis of LO-free detection:

$$|\Omega_{\text{RF}}| \longrightarrow \text{AT peak separation}.$$

5.5 LO-Free Noise Model

The total LO-free noise is modeled as

$$n_{\text{Ry}} \sim \mathcal{CN}(0, \sigma_{\text{Ry}}^2). \quad (40)$$

with

$$\sigma_{\text{Ry}}^2 = \sigma_{\text{UN}}^2 + \sigma_{\text{QPN}}^2 + \sigma_{\text{BGN}}^2. \quad (41)$$

The three modeled contributions are:

1. background noise,
2. quantum projection noise,
3. observation uncertainty.

5.6 Quantum Projection Noise

The paper models the QPN approximately as

$$\sigma_{\text{QPN}}^2 \approx \frac{1}{4C_{\text{eff}}N_{\text{atoms}}}. \quad (42)$$

Here C_{eff} is the effective readout contrast/efficiency and N_{atoms} is the number of quantum-mechanically uncorrelated atoms.

5.7 Observation Uncertainty

The observation uncertainty is modeled as

$$\sigma_{\text{UN}}^2 = (\tilde{\epsilon}E_{\text{RF}})^2 = \tilde{\epsilon}^2|E_{\text{RF}}|^2. \quad (43)$$

Using $P_{\text{Rx}} = |E_{\text{RF}}|^2$,

$$\sigma_{\text{UN}}^2 = \tilde{\epsilon}^2P_{\text{Rx}}. \quad (44)$$

The reference value is

$$\tilde{\epsilon} = 0.5\%. \quad (45)$$

The paper also notes that observation uncertainty can increase when the RF field is attenuated and the AT peaks become more difficult to resolve.

5.8 LO-Free SNR

From the receiver model,

$$\gamma_{\text{LF}} = SNR_{\text{Ry}} = \frac{P_{\text{Rx}}|h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2P_{\text{Rx}}}. \quad (46)$$

Equivalently,

$$\gamma_{\text{LF}} = \frac{E_{\text{RF}}^2|h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2E_{\text{RF}}^2}. \quad (47)$$

The transition dipole moment $\wp_{\text{RF}}/\hbar$ is a common multiplicative factor in the observation amplitude and therefore cancels from this SNR ratio when the noise is expressed in the same observation units.

5.9 LO-Free Modulation

The LO-free receiver measures amplitude and therefore uses PAM:

$$\boxed{\text{LO-free} \longrightarrow M\text{-PAM.}} \quad (48)$$

Using the PAM expression,

$$\boxed{SER_{\text{LF}} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\gamma_{\text{LF}}}{M^2-1}}\right).} \quad (49)$$

5.10 Complete Analytical LO-Free SER Equation

Substituting the LO-free SNR,

$$\boxed{SER_{\text{LF}} = \frac{2(M-1)}{M} Q\left[\sqrt{\frac{6}{M^2-1} \frac{P_{\text{Rx}}|h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}}\right].} \quad (50)$$

Using the QPN expression,

$$\boxed{SER_{\text{LF}} = \frac{2(M-1)}{M} Q\left[\sqrt{\frac{6}{M^2-1} \frac{P_{\text{Rx}}|h|^2}{\sigma_{\text{BGN}}^2 + \frac{1}{4C_{\text{eff}}N_{\text{atoms}}} + \tilde{\epsilon}^2 P_{\text{Rx}}}}\right].} \quad (51)$$

5.11 LO-Free SER Versus SNR

Inside the valid operating region, if the plotted SNR is directly identified with the Proakis SNR argument,

$$\boxed{\gamma = 10^{SNR_{\text{dB}}/10},} \quad (52)$$

and

$$\boxed{SER_{\text{LF}} = \frac{2(M-1)}{M} Q\left[\sqrt{\frac{6 \cdot 10^{SNR_{\text{dB}}/10}}{M^2-1}}\right].} \quad (53)$$

This is the classical PAM SER branch. It does not by itself describe the low-SNR distortion shown in the complete LO-free Fig. 9 curve.

5.12 LO-Free Distortion Region

The quantum receiver introduces an additional operating condition:

$$\boxed{|\Omega_{\text{RF}}| \geq \Omega_{\text{RF, th}} \implies \text{AT-splitting operating region.}} \quad (54)$$

Conversely,

$$\boxed{|\Omega_{\text{RF}}| < \Omega_{\text{RF, th}} \implies \text{LO-free distortion region.}} \quad (55)$$

The paper's numerical characterization shows a transition around

$$\frac{\Omega_{\text{RF}}}{2\pi} \approx 5.5 \text{ MHz}$$

for the particular quantum configuration shown in the distortion study. This threshold is configuration-dependent and should not be treated as a universal constant.

5.13 Piecewise LO-Free SER Model

The complete communication model is therefore

$$SER_{\text{LF}} = \begin{cases} \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\gamma_{\text{LF}}}{M^2-1}}\right), & |\Omega_{\text{RF}}| \geq \Omega_{\text{RF, th}}, \\ SER_{\text{distortion, LF}}, & |\Omega_{\text{RF}}| < \Omega_{\text{RF, th}}. \end{cases} \quad (56)$$

The first branch is analytically obtained from the receiver SNR and Proakis PAM SER. The second branch is determined by the quantum AT-splitting response and is not supplied by the paper as a universal closed-form SER expression.

5.14 Complete LO-Free Mathematical Chain

$$SNR \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}| \quad (57)$$

followed by

$$|\Omega_{\text{RF}}| \longrightarrow \begin{cases} \text{AT operating region,} \\ \text{AT distortion region.} \end{cases} \quad (58)$$

Inside the operating region,

$$E_{\text{RF}} \longrightarrow \sigma_{\text{Ry}}^2 \longrightarrow \gamma_{\text{LF}} \longrightarrow SER_{\text{PAM}}. \quad (59)$$

6 PART C – LO-Dressed Rydberg Atomic Receiver

6.1 LO-Dressed Receiver Principle

The LO-dressed receiver uses an RF signal and a local oscillator.

The RF field is

$$E_{\text{RF}}(t) = A_{\text{RF}} \cos(2\pi f_{\text{RF}} t + \phi_{\text{RF}}). \quad (60)$$

The LO field is

$$E_{\text{LO}}(t) = A_{\text{LO}} \cos(2\pi f_{\text{LO}} t + \phi_{\text{LO}}). \quad (61)$$

The frequency and phase differences are

$$\Delta f = f_{\text{LO}} - f_{\text{RF}}, \quad (62)$$

$$\Delta \phi = \phi_{\text{LO}} - \phi_{\text{RF}}. \quad (63)$$

The total field is

$$E_{\text{total}} = E_{\text{LO}} + E_{\text{RF}}. \quad (64)$$

For

$$\frac{A_{\text{RF}}}{A_{\text{LO}}} \ll 1,$$

the rotating-wave/small-signal representation becomes

$$E_{\text{total}} \approx [A_{\text{LO}} + A_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)] \cos(2\pi f_{\text{LO}} t + \phi_{\text{LO}}). \quad (65)$$

6.2 LO-Dressed Atomic Response

For implementation, the probe-transmission model is first written in its explicit form. The adiabatic probe transmission is

$$P_{\text{out}}(t) = P_{\text{in}} \exp[-k_p L \text{Im}\{\chi(t)\}], \quad (\text{P2})$$

where

$$P_{\text{in}} = \frac{\pi}{2\eta_0} \left(\frac{d\Omega_p \hbar}{2\sqrt{2} \wp_{12}} \right)^2, \quad (\text{P3})$$

and

$$\chi(t) = C_0 \rho_{21}(t), \quad C_0 = -\frac{2N_0 \wp_{12}}{\epsilon_0 \hbar \Omega_p}. \quad (\text{P4})$$

Here $k_p = 2\pi/\lambda_p$, L is the vapor-cell length, d is the $1/e^2$ probe-beam diameter, η_0 is the free-space impedance, N_0 is the atomic density, and ρ_{21} is the steady-state density-matrix component of the $|1\rangle \leftrightarrow |2\rangle$ transition.

For the LO-dressed small-signal model, define

$$\bar{A} = \frac{\gamma_2 \Omega_p}{\gamma_2^2 + 2\Omega_p^2}, \quad (\text{P5})$$

and the effective probe absorption coefficient

$$\alpha = k_p L C_0 \bar{A}. \quad (\text{P6})$$

The DC probe output power is then

$$\bar{P}_0 = P_{\text{in}} e^{-\alpha} e^{\alpha \Lambda(\Omega_{\text{LO}}, \Gamma)}. \quad (\text{P7})$$

The overall atomic response slope is

$$\kappa = \alpha \bar{P}_0 \kappa_\rho. \quad (\text{P8})$$

The tags (P2)–(P8) identify the source-paper equations/definitions and do not alter the equation numbering of this document.

The supplementary derivation introduces the normalized Lorentzian

$$\Lambda(a, b) = \frac{b^2}{b^2 + a^2}. \quad (66)$$

The intrinsic atomic-response slope is

$$\kappa_\rho = \frac{\partial \Lambda(\Omega_{\text{LO}}, \Gamma)}{\partial \Omega_{\text{LO}}}. \quad (67)$$

The EIT half-width quantities are

$$\Gamma_{\text{EIT}} = \frac{\Omega_c^2 + \Omega_p^2}{2\sqrt{\gamma_2^2 + 2\Omega_p^2}}, \quad (68)$$

and

$$\Gamma = \frac{2\sqrt{2}\Omega_p}{\sqrt{\Omega_c^2 + \Omega_p^2}} \Gamma_{\text{EIT}}. \quad (69)$$

The intrinsic optimum is obtained from

$$\frac{\partial^2 \Lambda(\Omega_{\text{LO}}, \Gamma)}{\partial \Omega_{\text{LO}}^2} = 0, \quad (70)$$

giving

$$\Omega_{\text{LO}} = \frac{\sqrt{3}}{3} \Gamma. \quad (71)$$

When the practical probe-transmission factor is included, the optimum local-oscillator Rabi frequency can be adjusted from the intrinsic value. The supplementary derivation gives

$$\Omega_{\text{LO}}^{(\text{opt})} = \left[\bar{A} - 1 + \sqrt{\bar{A}^2 - 2\bar{A} + 4} \right]^{1/2} \frac{\sqrt{3}}{3} \Gamma, \quad (72)$$

where $\bar{A}$ is the zero-RF EIT response parameter used in the supplementary derivation. The intrinsic optimum is the simpler form obtained directly from the Lorentzian slope.

The corresponding maximum intrinsic slope is

$$(\kappa_\rho)_{\text{max}} = \frac{3\sqrt{3}}{8\Gamma}. \quad (73)$$

The practical probe transmission before the small-signal expansion is represented as

$$P_{\text{out}}(t) = \bar{P}_0 e^{\alpha \kappa_\rho \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)}. \quad (74)$$

The practical probe transmission is represented in the small-signal linear region by

$$P_{\text{out}}(t) \approx \bar{P}_0 + \kappa \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi). \quad (75)$$

Here $\bar{P}_0$ is the DC probe output power and κ is the overall atomic response slope.

The full small-signal condition is

$$|\kappa_\rho \Omega_{\text{RF}}| \ll 1, \quad (76)$$

together with the physical requirement

$$|\Omega_{\text{RF}}| \ll |\Omega_{\text{LO}}|$$

for the RF perturbation to remain small relative to the LO dressing.

6.3 RF Rabi Frequency

As in the LO-free receiver,

$$\Omega_{\text{RF}} = \frac{\wp_{\text{RF}} |E_{\text{RF}}|}{\hbar}. \quad (77)$$

Hence,

$$\Omega_{\text{RF}} = \frac{\wp_{\text{RF}}}{\hbar} \sqrt{P_{\text{Rx}}} \quad (78)$$

up to the field-sign convention.

6.4 Amplitude and Phase Recovery

The amplitude of the Fourier component at Δf satisfies

$$\Omega_{\text{RF}} = \frac{|P(\Delta f)|}{|\kappa|}. \quad (79)$$

The LO provides a phase reference, so both RF amplitude and phase can be recovered. Consequently,

$$\text{LO-dressed} \longrightarrow M\text{-QAM}. \quad (80)$$

6.5 Photodetection

The AC photocurrent is

$$I_{\text{AC}} = D|\kappa|\Omega_{\text{RF}}. \quad (81)$$

Substituting the RF Rabi-frequency relation,

$$I_{\text{AC}} = D|\kappa|\frac{\wp_{\text{RF}}}{\hbar}E_{\text{RF}}. \quad (82)$$

Using $|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}$,

$$I_{\text{AC}} = D|\kappa|\frac{\wp_{\text{RF}}}{\hbar}\sqrt{P_{\text{Rx}}}. \quad (83)$$

6.6 Photocurrent Power

The electrical power associated with the AC photocurrent is

$$P_{e,\text{AC}} = I_{\text{AC}}^2 R_L. \quad (84)$$

Therefore,

$$P_{e,\text{AC}} = R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} P_{\text{Rx}}. \quad (85)$$

Thus the useful LO-dressed signal power is proportional to P_{Rx} .

6.7 LO-Dressed Communication Signal Model

The observation model is

$$z_{\text{LO}} = \sqrt{P_{\text{Rx}} G_{\text{LNA}} R_L D^2} |\kappa| \frac{\wp_{\text{RF}}}{\hbar} h x + n_{\text{Ry,LO}}. \quad (86)$$

Equivalently, defining

$$K_{\text{LO}} = \sqrt{G_{\text{LNA}} R_L D^2} |\kappa| \frac{\wp_{\text{RF}}}{\hbar}, \quad (87)$$

we have

$$z_{\text{LO}} = K_{\text{LO}} \sqrt{P_{\text{Rx}}} h x + n_{\text{Ry,LO}}. \quad (88)$$

6.8 LO-Dressed Noise Sources

The modeled LO-dressed noise sources are:

1. background noise,
2. photon shot noise,
3. thermal noise.

The QPN contribution is neglected in the LO-dressed model. Thus,

$$n_{\text{Ry,LO}} \sim \mathcal{CN}(0, \sigma_{\text{Ry,LO}}^2). \quad (89)$$

6.9 Background Noise Contribution

The paper models the Fourier-component amplitude in the presence of background noise as

$$|P(\Delta f)| = |\kappa| \frac{\mathcal{P}_{\text{RF}}}{\hbar} \left(\sqrt{P_{\text{Rx}}} + n_{\text{BGN}} \right). \quad (90)$$

After photodetection and LNA processing, the background-noise contribution to the electrical noise variance is

$$\sigma_{\text{BGN,LO}}^2 = G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\mathcal{P}_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2. \quad (91)$$

6.10 Photon Shot Noise

The photon shot-noise variance is

$$\sigma_{\text{PSN}}^2 = 2eB\bar{I}. \quad (92)$$

For the numerical model, the average optical power reaching the photodiode is explicitly identified with the same DC probe output power $\bar{P}_0$ introduced in Eq. (P7):

$$\bar{P}_{\text{opt}} = \bar{P}_0. \quad (93)$$

Hence the average photocurrent is

$$\bar{I} = \eta_{\text{eff}} \frac{\bar{P}_0}{\hbar\omega_p} e, \quad \omega_p = 2\pi f_p. \quad (94)$$

Therefore,

$$\sigma_{\text{PSN}}^2 = \frac{2\eta_{\text{eff}} e^2 B \bar{P}_0}{\hbar\omega_p}. \quad (95)$$

This explicitly ties the photon-flux calculation to the already-derived $\bar{P}_0$, rather than using the instantaneous communication sideband $|P(\Delta f)|$ as the average optical power.

6.11 Thermal Noise

The thermal-noise variance is

$$\sigma_{\text{TN}}^2 = 4k_B T B. \quad (96)$$

6.12 Photocurrent and LNA Output Power

The complete electrical AC power relation is

$$\begin{aligned} P_{e,\text{AC}} = & R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} P_{\text{Rx}} + R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2 \\ & + R_L D^2 \sigma_{\text{PSN}}^2. \end{aligned} \quad (97)$$

After the LNA,

$$\begin{aligned} P_{\text{LNA}} = & G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} P_{\text{Rx}} \\ & + G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2 \\ & + G_{\text{LNA}} R_L D^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \end{aligned} \quad (98)$$

The first term is the useful signal power and the remaining terms form the modeled noise variance:

$$\begin{aligned} \sigma_{\text{Ry,LO}}^2 = & G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2 \\ & + G_{\text{LNA}} R_L D^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \end{aligned} \quad (99)$$

6.13 LO-Dressed SNR

The useful signal power is

$$P_{\text{signal,LO}} = G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} P_{\text{Rx}} |h|^2. \quad (100)$$

Hence,

$$\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}^2}{\hbar^2} \right) P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (101)$$

6.14 LO-Dressed QAM SER

The LO-dressed receiver supports amplitude and phase:

$$\text{LO-dressed} \longrightarrow M\text{-QAM}. \quad (102)$$

Therefore,

$$SER_{\text{LO}} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3\gamma_{\text{LO}}}{M-1}} \right) \right]^2. \quad (103)$$

6.15 Complete Analytical LO-Dressed SER Equation

Substitution of the LO-dressed SNR into the QAM expression gives

$$SER_{LO} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3}{M-1} \frac{G_{LNA} R_L D^2 \kappa^2 \left(\frac{\wp_{RF}^2}{\hbar^2} \right) P_{Rx} |h|^2}{\sigma_{Ry,LO}^2}} \right) \right]^2. \quad (104)$$

Using the total LO-dressed noise variance,

$$SER_{LO} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3}{M-1} \frac{G_{LNA} R_L D^2 \kappa^2 \left(\frac{\wp_{RF}^2}{\hbar^2} \right) P_{Rx} |h|^2}{G_{LNA} R_L D^2 \kappa^2 \left(\frac{\wp_{RF}^2}{\hbar^2} \right) \sigma_{BGN}^2 + G_{LNA} R_L D^2 \sigma_{PSN}^2 + \sigma_{TN}^2}} \right) \right]^2. \quad (105)$$

6.16 LO-Dressed Linear Dynamic Range

The analytical LO-dressed SNR and SER model assumes

$$P_{out} \propto \Omega_{RF}.$$

This relation is valid only inside the approximately linear dynamic range.

Define the upper RF-Rabi-frequency boundary as

$$|\Omega_{RF}| \leq \Omega_{RF, \max} \implies \text{linear operating region.} \quad (106)$$

For

$$|\Omega_{RF}| > \Omega_{RF, \max}, \quad (107)$$

the atomic response becomes nonlinear and distortion appears.

The exact boundary is configuration-dependent and is obtained from the quantum response rather than from a universal closed-form equation.

6.17 Why QuTiP Is Required

The atomic master equation determines the nonlinear probe transmission through the total Rabi frequency. Because the exact linear-dynamic-range boundary depends on the quantum-system parameters, a numerical quantum simulation is used to evaluate

$$P_{out}(\Omega_{total})$$

and identify the approximately linear interval.

Thus,

$$\text{master equation} \longrightarrow \text{QuTiP} \longrightarrow P_{out}(\Omega_{total}) \longrightarrow \text{linear dynamic range.} \quad (108)$$

6.18 High-SNR Distortion Mechanism

At moderate signal strength,

$$SNR \uparrow \Rightarrow SER \downarrow.$$

However,

$$SNR \uparrow \Rightarrow P_{Rx} \uparrow \Rightarrow |E_{RF}| \uparrow \Rightarrow |\Omega_{RF}| \uparrow.$$

Eventually,

$$|\Omega_{RF}| > \Omega_{RF, \max},$$

so the atomic response becomes nonlinear.

Hence,

$$\boxed{\text{strong RF field} \longrightarrow \text{nonlinear atomic response} \longrightarrow \text{distortion.}} \quad (109)$$

Thus the practical LO-dressed SER may decrease initially and deteriorate again when the RF Rabi frequency becomes too large.

6.19 Piecewise LO-Dressed SER Model

The complete model is

$$SER_{\text{LO-dressed}} = \begin{cases} SER_{\text{QAM}}(\gamma_{\text{LO}}, M), & |\Omega_{\text{RF}}| \leq \Omega_{\text{RF}, \max}, \\ SER_{\text{distortion, LO}}, & |\Omega_{\text{RF}}| > \Omega_{\text{RF}, \max}. \end{cases} \quad (110)$$

The first branch is analytically obtained from the receiver SNR and Proakis QAM equation. The second branch must be characterized from the nonlinear quantum response.

6.20 Complete LO-Dressed Mathematical Chain

$$\boxed{SNR \longrightarrow P_{Rx} \longrightarrow |E_{RF}| \longrightarrow |\Omega_{RF}|} \quad (111)$$

followed by

$$|\Omega_{\text{RF}}| \longrightarrow \begin{cases} \text{linear operating region,} \\ \text{nonlinear distortion region.} \end{cases} \quad (112)$$

Inside the linear region,

$$\boxed{\Omega_{\text{RF}} \longrightarrow P_{\text{out}} \longrightarrow I_{\text{AC}} \longrightarrow \sigma_{\text{Ry, LO}}^2 \longrightarrow \gamma_{\text{LO}} \longrightarrow SER_{\text{QAM}.}} \quad (113)$$

7 Comparison of the Three Receiver Models

7.1 Conventional RF Receiver

$$\boxed{y = \sqrt{P_c} h x + n_{\text{Conv}}, \quad n_{\text{Conv}} \sim \mathcal{CN}(0, \sigma_{\text{Conv}}^2).} \quad (114)$$

The literal conventional SNR is

$$\boxed{\gamma_{\text{Conv}} = \frac{P_c |h|^2}{G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B}.} \quad (115)$$

The modulation is QAM:

$$\boxed{SER_{\text{Conv}} = SER_{\text{QAM}}(\gamma_{\text{Conv}}).} \quad (116)$$

7.2 LO-Free Receiver

$$z = \frac{\wp_{\text{RF}}}{\hbar} \left| \sqrt{P_{\text{Rx}}} hx + n_{\text{Ry}} \right|. \quad (117)$$

The noise variance is

$$\sigma_{\text{Ry}}^2 = \sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}. \quad (118)$$

The SNR is

$$\gamma_{\text{LF}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}. \quad (119)$$

The modulation is PAM:

$$SER_{\text{LF}} = SER_{\text{PAM}}(\gamma_{\text{LF}}). \quad (120)$$

The additional quantum condition is

$$|\Omega_{\text{RF}}| \geq \Omega_{\text{RF}, \text{th}}. \quad (121)$$

7.3 LO-Dressed Receiver

The signal model is

$$z_{\text{LO}} = \sqrt{P_{\text{Rx}} G_{\text{LNA}} R_L D^2} |\kappa| \frac{\wp_{\text{RF}}}{\hbar} hx + n_{\text{Ry,LO}}. \quad (122)$$

The noise variance is

$$\begin{aligned} \sigma_{\text{Ry,LO}}^2 = & G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2 \\ & + G_{\text{LNA}} R_L D^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \end{aligned} \quad (123)$$

The SNR is

$$\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}^2}{\hbar^2} \right) P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (124)$$

The modulation is QAM:

$$SER_{\text{LO}} = SER_{\text{QAM}}(\gamma_{\text{LO}}). \quad (125)$$

The additional quantum condition is

$$|\Omega_{\text{RF}}| \leq \Omega_{\text{RF}, \text{max}}. \quad (126)$$

8 Final Mathematical Relationship for Fig. 9

The common physical chain is

$$\boxed{\text{RF signal} \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}|.} \quad (127)$$

For the conventional receiver,

$$\boxed{\text{Conventional:} \quad \text{SNR} \longrightarrow \text{SER}_{\text{QAM}}.} \quad (128)$$

For the LO-free receiver,

$$\boxed{\text{LO-free:} \quad \text{SNR} \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}| \longrightarrow \text{AT condition} \longrightarrow \text{SER}_{\text{PAM}}.} \quad (129)$$

For the LO-dressed receiver,

$$\boxed{\text{LO-dressed:} \quad \text{SNR} \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}| \longrightarrow \text{linear/nonlinear condition} \longrightarrow \text{SER}_{\text{QAM}}.} \quad (130)$$

The essential reconstruction principle is

$$\boxed{\text{quantum receiver model} \longrightarrow \text{effective SNR} \longrightarrow \text{Proakis SER}.} \quad (131)$$

9 Important Interpretation of Fig. 9

9.1 Conventional Curve

The conventional curve is monotonic:

$$\boxed{\text{SNR} \uparrow \Rightarrow \text{SER} \downarrow .} \quad (132)$$

It is an ordinary AWGN QAM SER relationship and does not require a quantum-system simulation for its basic analytical form.

9.2 LO-Free Curve

The LO-free receiver has a low-RF limitation:

$$P_{\text{Rx}} \downarrow \Rightarrow |E_{\text{RF}}| \downarrow \Rightarrow |\Omega_{\text{RF}}| \downarrow \Rightarrow \Delta\nu_{\text{AT}} \downarrow .$$

Eventually the AT peaks become difficult to resolve:

$$\boxed{\text{weak RF field} \longrightarrow \text{unresolved AT splitting} \longrightarrow \text{observation distortion}.} \quad (133)$$

Therefore a single textbook PAM curve cannot reproduce the entire LO-free distortion behavior without an additional quantum operating condition.

9.3 LO-Dressed Curve

The LO-dressed receiver has the opposite limitation:

$$P_{\text{Rx}} \uparrow \Rightarrow |E_{\text{RF}}| \uparrow \Rightarrow |\Omega_{\text{RF}}| \uparrow .$$

Eventually the atomic system leaves its linear dynamic range:

$$\boxed{\text{strong RF field} \longrightarrow \text{nonlinear atomic response} \longrightarrow \text{distortion}.} \quad (134)$$

Thus the practical LO-dressed SER may decrease initially and deteriorate again when the RF Rabi frequency becomes too large.

10 Why the Three Curves Are Not Simply Three Proakis Curves

The Proakis expressions establish

$$SNR \longrightarrow SER.$$

For the conventional receiver this is essentially the complete analytical relationship.

For the Rydberg receivers, however, the effective SNR is generated by the receiver architecture:

$$\gamma_{LF} = f_{LF}(P_{Rx}, E_{RF}, \Omega_{RF}, \sigma_{BGN}, \sigma_{QPN}, \tilde{\epsilon}), \quad (135)$$

and

$$\gamma_{LO} = f_{LO}(P_{Rx}, \kappa, \wp_{RF}, G_{LNA}, R_L, D, \sigma_{BGN}, \sigma_{PSN}, \sigma_{TN}). \quad (136)$$

Therefore the complete Fig. 9 reconstruction requires

$$\text{receiver physics} \longrightarrow \text{effective SNR} \longrightarrow \text{Proakis SER}. \quad (137)$$

The analytical equations determine the SER inside the valid receiver operating regions. The non-monotonic distortion portions require the atomic response characterization.

11 Numerical Parameters From the Paper

11.1 Wireless-System Parameters

Table 1: Wireless-system parameters used in the reference study.

Parameter	Symbol	Value
Transmit antenna gain	G_{Tx}	2.15 dBi
Transmit power	P_{Tx}	0 dBm
Tx–Rx distance	d_{Tx-Rx}	10 m
Background-noise power	σ_{BGN}^2	−90 dBm
Receiver antenna gain	G_{Rx}	2.15 dBi
RF carrier frequency	f_{RF}	3.5 GHz
Bandwidth	B	1 MHz
Temperature	T	290 K
Free-space impedance	η_0	377 Ω
Vacuum permittivity	ϵ_0	8.854×10^{-12} F/m

For numerical work,

$$P_{dBm} \rightarrow P_W$$

must be performed using

$$P_W = 10^{(P_{dBm}-30)/10}. \quad (138)$$

A gain in dBi is converted using Eq. (5).

11.2 Atomic-System Parameters

The paper considers the four-level Cs system

$$\boxed{|1\rangle \rightarrow |2\rangle \rightarrow |3\rangle \rightarrow |4\rangle} \quad (139)$$

with

$$\boxed{6S_{1/2} \rightarrow 6P_{3/2} \rightarrow 47D_{5/2} \rightarrow 48P_{3/2}.} \quad (140)$$

Table 2: Atomic-system parameters reported for the quantum simulation.

Parameter	Symbol	Value
Vapor-cell length	L	1 cm
Atomic density	N_0	$1 \times 10^{15} \text{ m}^{-3}$
State- $ 1\rangle$ decay	γ_1	0
State- $ 2\rangle$ decay	$\gamma_2/(2\pi)$	5.2 MHz
State- $ 3\rangle$ decay	$\gamma_3/(2\pi)$	3.9 kHz
State- $ 4\rangle$ decay	$\gamma_4/(2\pi)$	1.7 kHz
RF transition dipole	$\wp_{\text{RF}}$	$-1443.459 ea_0$
Probe transition dipole	$\wp_{12}$	$(2.5 ea_0)^2$
RF carrier frequency	f_{RF}	3.5 GHz
Probe wavelength	λ_p	852 nm
Coupling wavelength	λ_c	510 nm
Probe Rabi frequency	$\Omega_p/(2\pi)$	8 MHz
Coupling Rabi frequency	$\Omega_c/(2\pi)$	1 MHz

The atomic density and other quantum parameters should be taken from the original PDF rendering when constructing the final QuTiP model.

11.3 LO-Free Parameters

The reference study uses

$$\boxed{\frac{\Omega_{\text{RF}}}{2\pi} = 6 \text{ MHz}} \quad (141)$$

for the LO-free numerical configuration.

The reference observation uncertainty is

$$\boxed{\tilde{\epsilon} = 0.5\%}. \quad (142)$$

The QPN model is

$$\boxed{\sigma_{\text{QPN}}^2 \approx \frac{1}{4C_{\text{eff}} N_{\text{atoms}}}}. \quad (143)$$

The complete LO-free noise variance is

$$\boxed{\sigma_{\text{Ry}}^2 = \sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}. \quad (144)$$

11.4 LO-Dressed Parameters

The reference configuration specifies

$$\boxed{\frac{\Omega_{\text{RF}}}{2\pi} = 20 \text{ kHz},} \quad (145)$$

$$\boxed{\frac{\Omega_{\text{LO}}}{2\pi} = 4.23 \text{ MHz},} \quad (146)$$

and

$$\boxed{\Delta f = 150 \text{ kHz}.} \quad (147)$$

Table 3: LO-dressed receiver parameters.

Parameter	Symbol	Value
LNA gain	G_{LNA}	20 dB
Load resistance	R_L	50 Ω
Photodiode responsivity	D	0.55 A/W
Temperature	T	290 K
Photon efficiency	η_{eff}	0.8
RF Rabi frequency	$\Omega_{\text{RF}}/2\pi$	20 kHz
LO Rabi frequency	$\Omega_{\text{LO}}/2\pi$	4.23 MHz
RF-LO frequency difference	Δf	150 kHz

The LNA gain in the equations must be converted from dB to linear gain:

$$\boxed{G_{\text{LNA,lin}} = 10^{20/10} = 100.} \quad (148)$$

12 Modulation Orders Used in Fig. 9

The modulation orders in Fig. 9 are represented by their binary order $\log_2 M$.

Table 4: Modulation orders corresponding to the Fig. 9 curves.

Receiver	$\log_2 M$ values	Modulation
Conventional	2, 4, 6, 8	Square QAM
LO-dressed	4, 6, 8, 10	Square QAM
LO-free	2, 4, 6, 8	PAM

For the QAM curves, the analytical expression in Eq. (12) is the square- QAM expression. The values $M = 4, 16, 64, 256$ satisfy this condition.

13 Constants Required for Numerical Simulation

Table 5: Physical constants.

Constant	Symbol	Value
Boltzmann constant	k_B	$1.380649 \times 10^{-23} \text{ J/K}$
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34} \text{ J s}$
Elementary charge	e	$1.602176634 \times 10^{-19} \text{ C}$
Bohr radius	a_0	$5.29177210903 \times 10^{-11} \text{ m}$
Vacuum permittivity	ϵ_0	$8.854 \times 10^{-12} \text{ F/m}$
Free-space impedance	η_0	$377 \text{ } \Omega$

The elementary charge must be used with the correct SI exponent:

$$e = 1.602176634 \times 10^{-19} \text{ C}.$$

The transition dipole moment is evaluated from

$$\boxed{\varphi_{\text{RF}} = -1443.459 ea_0.} \quad (149)$$

No absolute-value operator is inserted into the definition of φ_{RF} . When a power or variance is formed, the square naturally produces $\varphi_{\text{RF}}^2 > 0$.

14 Important Source-Verification Notes

14.1 Atomic Density

The atomic density used in the quantum simulation must be read from the original PDF rendering rather than from OCR text alone. The physically consistent value used for the present reference is

$$\boxed{N_0 = 1 \times 10^{15} \text{ m}^{-3}.$$

This value should be retained consistently in the QuTiP implementation after checking the original source.

14.2 Units of Rabi Frequency

All master-equation calculations use angular frequency:

$$\Omega_{\text{RF}} \quad [\text{rad/s}].$$

The experimentally plotted frequency is

$$\frac{\Omega_{\text{RF}}}{2\pi} \quad [\text{Hz}].$$

Therefore,

$$\boxed{\Omega_{\text{RF}} = 2\pi \left(\frac{\Omega_{\text{RF}}}{2\pi} \right).} \quad (150)$$

14.3 Transition Dipole Moment

The reference notation is

$$\wp_{\text{RF}} = -1443.459 ea_0.$$

The field magnitude is represented separately by

$$|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}.$$

Thus the Rabi-frequency relation is retained as

$$\Omega_{\text{RF}} = \wp_{\text{RF}} E_{\text{RF}} / \hbar,$$

with the physical magnitude used when comparing RF strengths.

15 Important SNR-Definition Note for Fig. 9

Two related but distinct concepts must be separated:

1. the physical receiver SNR obtained from the receiver noise model;
2. the E_s/N_0 argument used by the Proakis SER expression.

For the closed-form SNR-to-received-power mapping used in the Fig. 9 reconstruction, the observation uncertainty branch of the LO-free model is written as

$$\boxed{\gamma = \frac{P_r}{\sigma_B^2 + \epsilon^2 P_r}}, \quad (\text{SNR-LF})$$

where $P_r \equiv P_{\text{Rx}}$, $\sigma_B^2 \equiv \sigma_{\text{BGN}}^2$, and the unit-average channel convention is used. Solving directly for received power gives

$$\boxed{P_r = \frac{\gamma \sigma_B^2}{1 - \gamma \epsilon^2}, \quad \gamma < \frac{1}{\epsilon^2}}. \quad (\text{INV-LF})$$

The denominator shows the finite SNR ceiling of this simplified LO-free model:

$$\gamma_{\text{max}} = \frac{1}{\epsilon^2}.$$

For $\epsilon = 0.5\% = 0.005$, this ceiling is

$$10 \log_{10} \left(\frac{1}{0.005^2} \right) \approx 46.02 \text{ dB}.$$

For the LO-dressed receiver, after the photon-shot-noise correction above, $\bar{P}_0$ is independent of P_r in the adopted small-signal model. Consequently the LO-dressed SNR is linear in received RF power:

$$\boxed{\gamma_{\text{LO}} = \frac{A P_r}{B}}, \quad (\text{SNR-LO})$$

where

$$\boxed{A = G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}}{\hbar} \right)^2, \quad B = \sigma_{\text{Ry,LO}}^2}. \quad (\text{AB-LO})$$

Therefore the closed-form inversion is

$$\boxed{P_r = \frac{\gamma B}{A}}. \quad (\text{INV-LO})$$

No numerical root solver is required for either inversion.

The receiver models provide quantities such as

$$\gamma_{\text{Conv}} = \frac{P_c |h|^2}{G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B}, \quad (150)$$

$$\gamma_{\text{LF}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}, \quad (151)$$

and

$$\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\phi_{\text{RF}}^2}{\hbar^2} \right) P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (152)$$

For the closed-form Fig. 9 power mappings, the simplified LO-free observation-uncertainty branch and the linear LO-dressed branch are the ones inverted in Eqs. (INV-LF) and (INV-LO), respectively.

The Proakis equations use the normalized symbol-energy SNR

$$\gamma = \frac{E_s}{N_0}.$$

For the first analytical reconstruction of Fig. 9, the plotted SNR in dB is therefore mapped to

$$\boxed{\gamma = 10^{SNR_{\text{dB}}/10}}. \quad (153)$$

This is the appropriate mathematical bridge for the classical SER curves. However, the exact numerical-generation procedure used by the authors for the full Fig. 9 Rydberg curves is not completely specified in the paper. Therefore the non-monotonic distortion portions cannot be claimed to follow from the analytical SER equation alone.

16 What Can and Cannot Be Derived Analytically

16.1 Conventional Receiver

The conventional curve is analytically determined by

$$\boxed{SNR \longrightarrow SER_{\text{QAM}}}. \quad (154)$$

No quantum simulation is required for the basic curve.

16.2 LO-Free Receiver

Inside the valid AT-splitting region,

$$\boxed{SNR_{\text{LF}} \longrightarrow SER_{\text{PAM}}}. \quad (155)$$

However, the transition into the distortion region requires

$$\Omega_{\text{RF}} \longrightarrow \text{AT splitting} \longrightarrow \text{resolvability threshold}.$$

Hence the complete model is

$$\boxed{\text{analytical PAM SER} + \text{quantum-derived operating threshold}}. \quad (156)$$

16.3 LO-Dressed Receiver

Inside the linear dynamic range,

$$\boxed{SNR_{\text{LO}} \longrightarrow SER_{\text{QAM}}.} \quad (157)$$

The high-RF distortion region requires

$$\Omega_{\text{RF}} \longrightarrow P_{\text{out}}(\Omega_{\text{total}}) \longrightarrow \text{linear-dynamic-range boundary.}$$

Hence

$$\boxed{\text{analytical QAM SER} + \text{quantum-derived nonlinear response.}} \quad (158)$$

17 Final Master Equations for the Thesis

17.1 Conventional

$$\boxed{y = \sqrt{P_c} h x + n_{\text{Conv}}} \quad (159)$$

$$\boxed{n_{\text{Conv}} \sim \mathcal{CN}(0, \sigma_{\text{Conv}}^2)} \quad (160)$$

$$\boxed{P_c = \left[\frac{P_{\text{Tx}} G_{\text{Tx}}}{4\pi d_{\text{Tx-Rx}}^2} \right] \left[\frac{\lambda_{\text{RF}}^2}{4\pi} \right]} \quad (161)$$

$$\boxed{\sigma_{\text{Conv}}^2 = G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B} \quad (162)$$

$$\boxed{\gamma_{\text{Conv}} = \frac{P_c |h|^2}{G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4k_B T B}} \quad (163)$$

$$\boxed{SER_{\text{Conv}} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3\gamma_{\text{Conv}}}{M-1}} \right) \right]^2} \quad (164)$$

17.2 LO-Free

$$\boxed{z = \frac{\wp_{\text{RF}}}{\hbar} \left| \sqrt{P_{\text{Rx}}} h x + n_{\text{Ry}} \right|} \quad (165)$$

$$\boxed{\sigma_{\text{Ry}}^2 = \sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}} \quad (166)$$

$$\boxed{\sigma_{\text{QPN}}^2 \approx \frac{1}{4C_{\text{eff}} N_{\text{atoms}}}} \quad (167)$$

$$\boxed{\sigma_{\text{UN}}^2 = \tilde{\epsilon}^2 P_{\text{Rx}}} \quad (168)$$

$$\boxed{\gamma_{\text{LF}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}} \quad (169)$$

$$SER_{\text{LF}} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\gamma_{\text{LF}}}{M^2-1}}\right) \quad (170)$$

subject to

$$|\Omega_{\text{RF}}| \geq \Omega_{\text{RF}, \text{th.}} \quad (171)$$

17.3 LO-Dressed

$$P_{\text{out}}(t) \approx \bar{P}_0 + \kappa \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi) \quad (172)$$

$$I_{\text{AC}} = D|\kappa|\Omega_{\text{RF}} \quad (173)$$

$$\bar{I} = \eta_{\text{eff}} \frac{\bar{P}_0}{\hbar \omega_p} e \quad (174)$$

$$\sigma_{\text{PSN}}^2 = 2eB\bar{I} \quad (175)$$

$$\sigma_{\text{TN}}^2 = 4k_B T B \quad (176)$$

$$\begin{aligned} \sigma_{\text{Ry,LO}}^2 = & G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} \sigma_{\text{BGN}}^2 \\ & + G_{\text{LNA}} R_L D^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2 \end{aligned} \quad (177)$$

$$\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}^2}{\hbar^2}\right) P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2} \quad (178)$$

$$SER_{\text{LO}} = 1 - \left[1 - 2 \left(1 - \frac{1}{\sqrt{M}} \right) Q\left(\sqrt{\frac{3\gamma_{\text{LO}}}{M-1}}\right) \right]^2 \quad (179)$$

subject to

$$|\Omega_{\text{RF}}| \leq \Omega_{\text{RF}, \text{max.}} \quad (180)$$

18 Final Simulation Flow

The common SNR conversion is

$$SNR_{\text{dB}} \longrightarrow \gamma = 10^{SNR_{\text{dB}}/10}. \quad (181)$$

For the conventional receiver,

$$\gamma \longrightarrow SER_{\text{QAM}}. \quad (182)$$

For the LO-free receiver,

$$\gamma \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}| \longrightarrow \text{AT condition} \longrightarrow \gamma_{\text{LF}} \longrightarrow SER_{\text{PAM}}. \quad (183)$$

For the LO-dressed receiver,

$$\boxed{\gamma \longrightarrow P_{\text{Rx}} \longrightarrow |E_{\text{RF}}| \longrightarrow |\Omega_{\text{RF}}| \longrightarrow \text{linear dynamic range} \longrightarrow \gamma_{\text{LO}} \longrightarrow SER_{\text{QAM}}.} \quad (184)$$

The implementation should therefore be separated into two stages:

$$\boxed{\begin{array}{c} \text{Stage 1: classical SER} \\ \gamma \rightarrow SER_{\text{PAM/QAM}} \\ \\ \text{Stage 2: Rydberg receiver response} \\ P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{operating/distortion condition.} \end{array}}$$

19 Final Summary

The three receiver models are summarized by

Table 6: Mathematical summary of the three receiver architectures.	
Receiver	Mathematical relationship
Conventional	$SNR \rightarrow \gamma_{\text{Conv}} \rightarrow SER_{\text{QAM}}$
LO-free	$SNR \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{AT condition} \rightarrow \gamma_{\text{LF}} \rightarrow SER_{\text{PAM}}$
LO-dressed	$SNR \rightarrow P_{\text{Rx}} \rightarrow E_{\text{RF}} \rightarrow \Omega_{\text{RF}} \rightarrow \text{linear/nonlinear condition} \rightarrow \gamma_{\text{LO}} \rightarrow SER_{\text{QAM}}$

The two central communication-theory expressions are

$$\boxed{SER_{\text{PAM}} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6\gamma}{M^2-1}}\right)} \quad (185)$$

and

$$\boxed{SER_{\text{QAM}} = 1 - \left[1 - 2\left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3\gamma}{M-1}}\right)\right]^2.} \quad (186)$$

The principal Rydberg relation is

$$\boxed{\Omega_{\text{RF}} = \frac{\wp_{\text{RF}} E_{\text{RF}}}{\hbar}.} \quad (187)$$

The LO-free SNR is

$$\boxed{\gamma_{\text{LF}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{BGN}}^2 + \sigma_{\text{QPN}}^2 + \tilde{\epsilon}^2 P_{\text{Rx}}}.} \quad (188)$$

The LO-dressed SNR is

$$\boxed{\gamma_{\text{LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}^2}{\hbar^2}\right) P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry,LO}}^2}.} \quad (189)$$

Finally,

$$\boxed{\text{Rydberg receiver response} \longrightarrow \text{effective SNR} \longrightarrow \text{Proakis SER.}} \quad (190)$$

The analytical SER equations determine the communication-theory performance within the valid operating region. The low-RF LO-free distortion and high-RF LO-dressed distortion require the corresponding quantum response characterization.